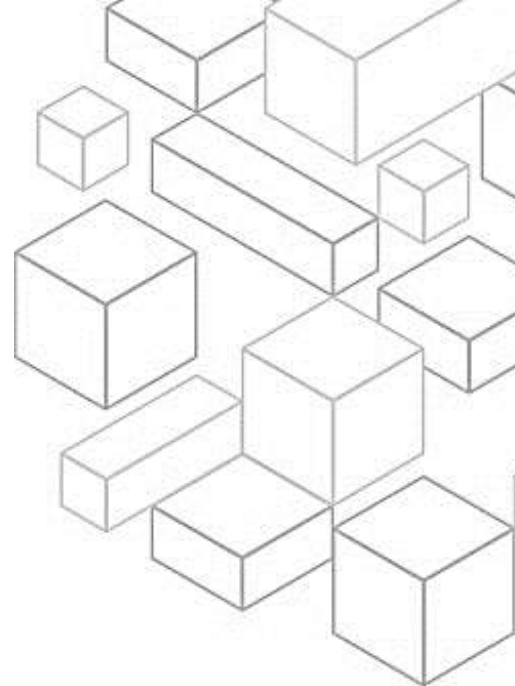




Transcript for AWS Well-Architected-Module2-WAFR

1.1 AWS Well-Architected

Welcome to module two of AWS Well-Architected: How to Run a Well-Architected Framework Review.

1.2 Learning objectives

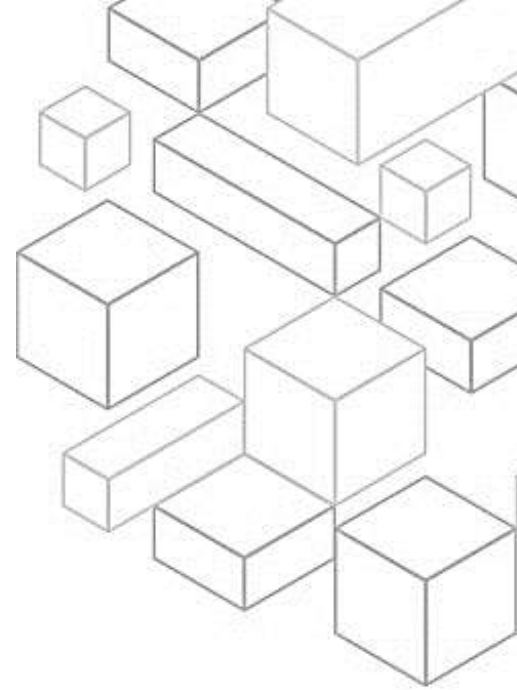
In this module, you will learn how to complete a Well-Architected Framework Review, understand the impacts of design decisions about your architecture, and evaluate risks in your architecture and how to mitigate them.

1.3 What Is the Well-Architected Framework Review?

The Well-Architected Framework Review is a continuous improvement mechanism that helps customers consistently evaluate their workloads against Amazon Web Services, or AWS, best practices. They can identify recommended remediations to address high-risk and medium-risk issues. The purpose of reviewing an architecture is to help identify any critical issues that might need to be addressed or areas that could be improved. The outcome of the review is a set of actions designed to improve the workload's architecture based on the six pillars of the framework.

1.4 A mechanism for continuous improvement

To achieve the desired goal from a framework review, it's important to consider it as a step in a continuous improvement plan that integrates with your workload lifecycle. This mechanism has three steps: learn, measure, and improve.



First, start with learning the strategies and best practices for architecting in the cloud.

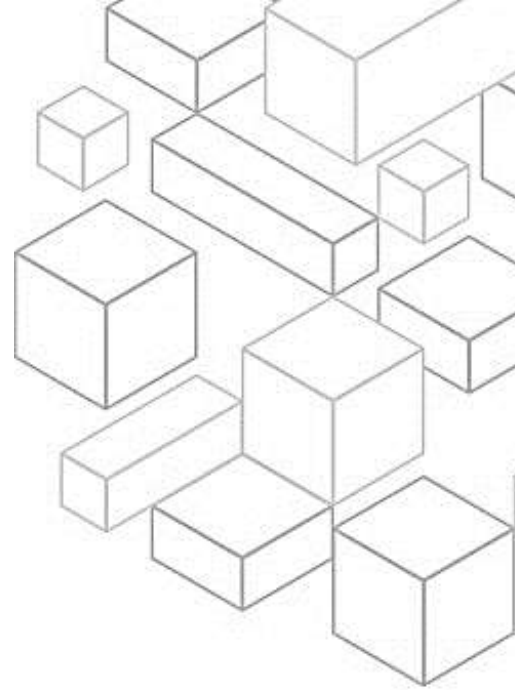
Next, you can measure your architecture using the framework; Well-Architected lenses, such as the data analytics lens; and your organization's best practices with the custom lenses in the AWS Well-Architected Tool, or AWS WA Tool.

Finally, you can use the outcome to improve your cloud architecture by addressing any high-risk issues. You can identify issues using improvement plans, Well-Architected Labs, the AWS Partner Network (APN), AWS solutions architecture teams, and more. You have to apply this three-step mechanism on every workload in your organization. A workload identifies a set of components that together deliver business value. You will learn more about workload details in a later module.

1.5 Intent of a review

The purpose of reviewing an architecture is to identify any critical issues that might need to be addressed or areas that could be improved. The outcome of the review is a set of actions that should improve the experience of using the workload. To achieve that goal, the review of architecture needs to be done consistently and with a blame-free approach that encourages the team to dive deep. It should be a lightweight process that is completed in hours, not days. This is a conversation, not an audit.

Team members who build an architecture using this framework should continually review their architecture, rather than hold a formal review meeting. A continuous approach helps your team members update answers as the architecture evolves, improving the architecture as you deliver features.



1.6 Learnings

What have we learned from doing reviews? Review early in the lifecycle when it is faster to fix things and can influence design. The most common issue is not making bad decisions but when decisions are neglected. Team members don't decide to not back up data; they just forget to talk about it. Most workloads have high-risk items that need to be addressed. Finding them is not a bad thing; they were always there. If you address them, it becomes one less thing that can damage or slow your business.

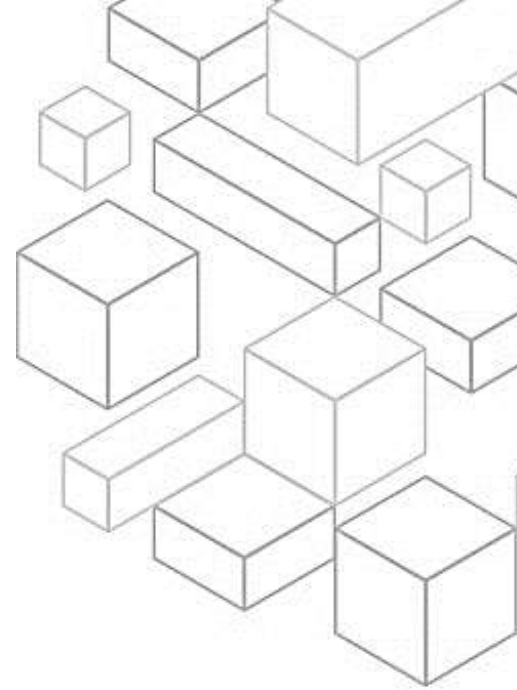
1.7 Use cases

Now, you will learn about some of the most common use cases. The first use case is learning best practices for the cloud. This applies to the majority of customers and their teams who want to learn how to architect for the cloud. By learning AWS best practices, companies can identify risks and improvement opportunity for their architecture. Technology governance is another consideration. Before launching into production, you want to know that you and your workload are ready. With lots of teams, it can be hard to know if they are all doing the right things. With any launch process or review, how can consistency be achieved? How are problems prioritized over time?

The AWS WA Tool provides a consistent process for measuring your architecture using AWS best practices. For portfolio management, most organizations are dependent on their technology portfolio to operate. Organizations often don't have a central registry of what is in that portfolio and what the risks are. This means that it can be difficult to make informed decisions about where to invest. Those organizations that do have a review process probably don't have a consistent, comprehensive one, and the results are not discoverable. By using the AWS WA Tool, you end up with a portfolio of workloads in your organization. You have a place to record metadata about the workloads, such as production, account, or Regions.



Customers often use the tool to document the architecture decisions they made. It creates a central view across all six pillars and any risks that exist. Senior management can then check for trends across the portfolio and any training that might be needed.



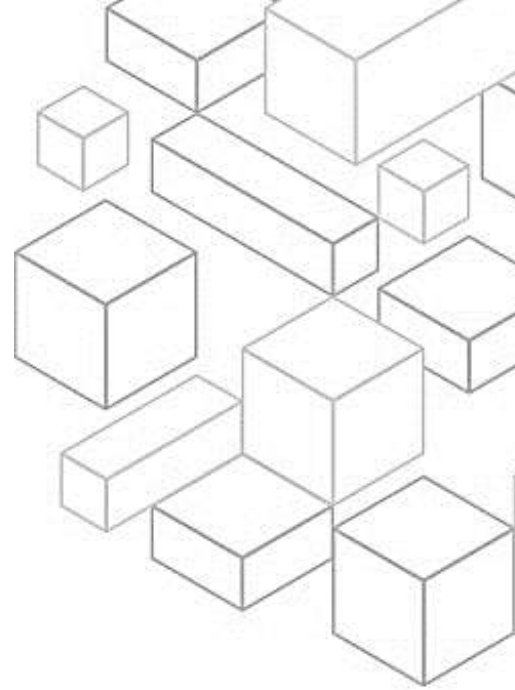
1.8 Three review phases

There are three phases to running a Well-Architected Framework

Review: prepare, review, and improve. In the prepare phase, you define a workload for review in your organization and identify individuals who can answer questions during the review on each pillar. You also need to identify someone to own the improvement plan and, eventually, the implementation of the improvement as a result of the review. These individuals are called sponsors. During the review phase, you run the actual review using the AWS WA Tool. Then, you publish the report that contains details about the current state of the workload, including notes and recommended improvement actions. During this phase, you identify high-risk issues and medium-risk issues for remediation. In the improve phase, you start reviewing the risk issues identified as part of the review, prioritize them, and create a detailed treatment plan to address them. You will dive deeper into each phase and best practices in a future module of this course.

1.9 Prepare

First, you will learn more about the prepare phase of the review.



1.10 Best practices for preparing reviews

What steps can you take to help prepare for the review? First, define the workload that will be reviewed. A workload can be a process, a technology, an infrastructure, a team, or a combination of them all, that deliver business value to your organization. For example, a website where you receive purchase orders from your customers can be a workload.

Next, identify the core team for the workload reviewed. This team is responsible for the success of that workload and contains subject matter experts for each pillar. Experts on this team should be able to answer the questions on each pillar and should own the future improvement plan that can result from identifying risks in the architecture. Then you need to hold a scoping session for the workload to be reviewed.

1.11 Best practices for preparing reviews (cont.)

Other preparations include deciding on the type of review. For example, is it a one-day session to review the six pillars, or multiple sessions to review pillars separately? Participants also need to prepare and gather necessary data to answer the review questions. Finally, you are ready to schedule the review.

1.12 Review preparation steps

The following is an example timeline that can help you plan for the review.

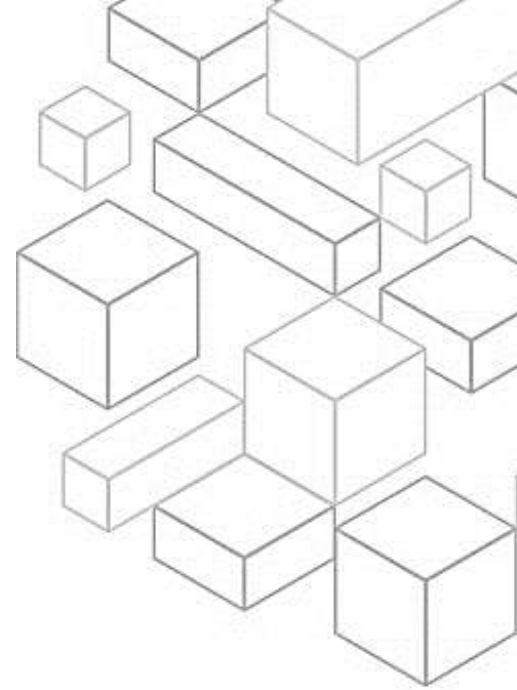
Approximately three weeks before a planned review, select a workload and core review team. Invite participants to the scoping meeting.

Around 16 days before the review, hold the scoping session. In this meeting, confirm and capture workload definitions. Select the appropriate questions from the AWS WA



Tool, including relevant lenses, where applicable. Identify subject matter experts for all relevant questions, and define the review type and approach. Request that participants gather relevant data that is accessible, rather than building new data for the review.

Then, five days before the review, the scope should be confirmed by all participants. Send a reminder for them to bring all relevant information that is readily available.



1.13 Review

The second phase is the review phase for conducting the actual review.

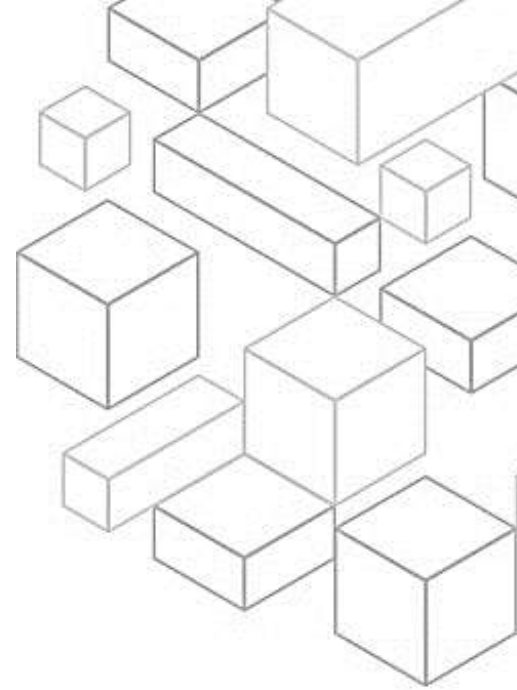
1.14 Best practices for running reviews

When conducting a review, some best practices are recommended.

- From the core review team, the moderator manages the meeting and sticks to the scope defined during preparation.
- It is helpful if the review team appoints one person to take notes on the discussion, and enters this into the tool. This role can be rotated in the team to spread the load and effort. It is important that the moderator is not also the note taker because this can lead to less productive reviews.
- Only one person updates the tool at a time.
- Fields do not dynamically change as modified across multiple clients. Updated records are overwritten.
- Use the AWS WA Tool to track results.
- One tip is to set up an Amazon Simple Storage Service, or Amazon S3, bucket to store analyses or other diagrams or documentation related to this workload.



- You can use a simple naming convention such as the account and workload name.



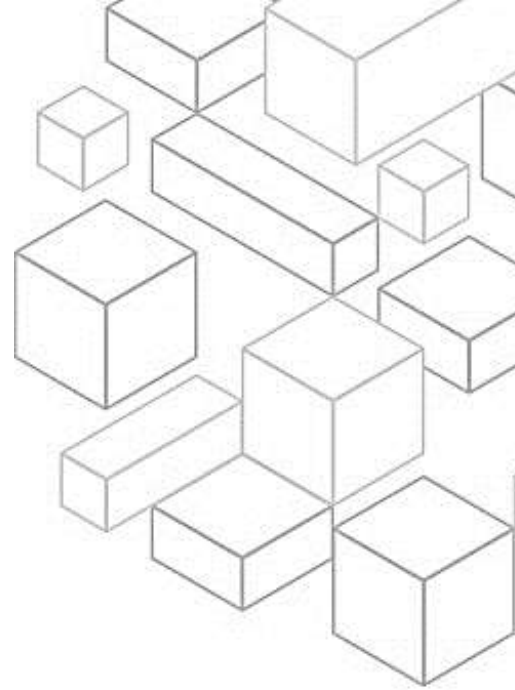
1.15 Improve

The third phase is the improve phase. This phase is about making an improvement plan to mitigate some of the risks identified as a result of conducting the review.

1.16 Risk prioritization methodology and considerations

The first step in the improve phase of the review is prioritizing risks. Before jumping into risk prioritization, it's a good idea to briefly define the term. Risk prioritizing is the process of identifying the most critical risks so they can be addressed first. Priorities should be set using the likelihood of a risk and the potential impact it poses to the organization or business. The aim is to determine a most-to-least-critical rank order of identified risks. Examples of risk impacts include lost sales, corporate liability, brand reputation damage, loss of market share, longer time to market, legal, regulatory, and so on.

Risks are identified based on business goals, needs, and priorities. Examples include time to market, security, reliability, performance, and cost. In the Well-Architected Framework, the risk levels are categorized as high or medium. As the name suggests, high-risk issues are architectural and operational choices that AWS has found might result in significant negative impact to a business. While medium risk issues also might negatively impact business, they typically do so to a lesser extent.



1.17 Well-Architected improvement workflow

Now that you know the risk levels, it is helpful to go over the improvement workflow to identify, prioritize, and later address these risks. Start with identifying the risks and improvement opportunities. This is achieved by running a Well-Architected Framework Review against a workload to understand where the workload is measured against cloud best practices in the framework. As you capture data about the workload, we generate insights to understand where those risks are and improvement opportunities.

Next, you will take the improvement opportunities and determine prescriptive solutions. These address the issues that take the greatest priority based on potential impact and the level of effort required to implement them and eliminate as many risks as possible at the same time. After solutions have been determined, it is important to identify which solutions take the greatest priority from the business perspective. Work can then begin on implementing the improvement solutions in order of priority. Improvement progress should be tracked, monitoring results to ensure that the desired benefits are achieved.

Best practices in the framework include guidance on people, processes, and technology. Implementing a missing best practice requires a combination of a collaboration between your AWS account team and actions on your part to apply them. These phases are explained in more detail in future modules.

1.22 Summary

In this module, you learned how to complete a Well-Architected Framework Review and the impacts of design decisions on your architecture. You also learned how to identify and evaluate risks in your architecture and how to mitigate them



1.23 Thank you

Thank you for your participation!

